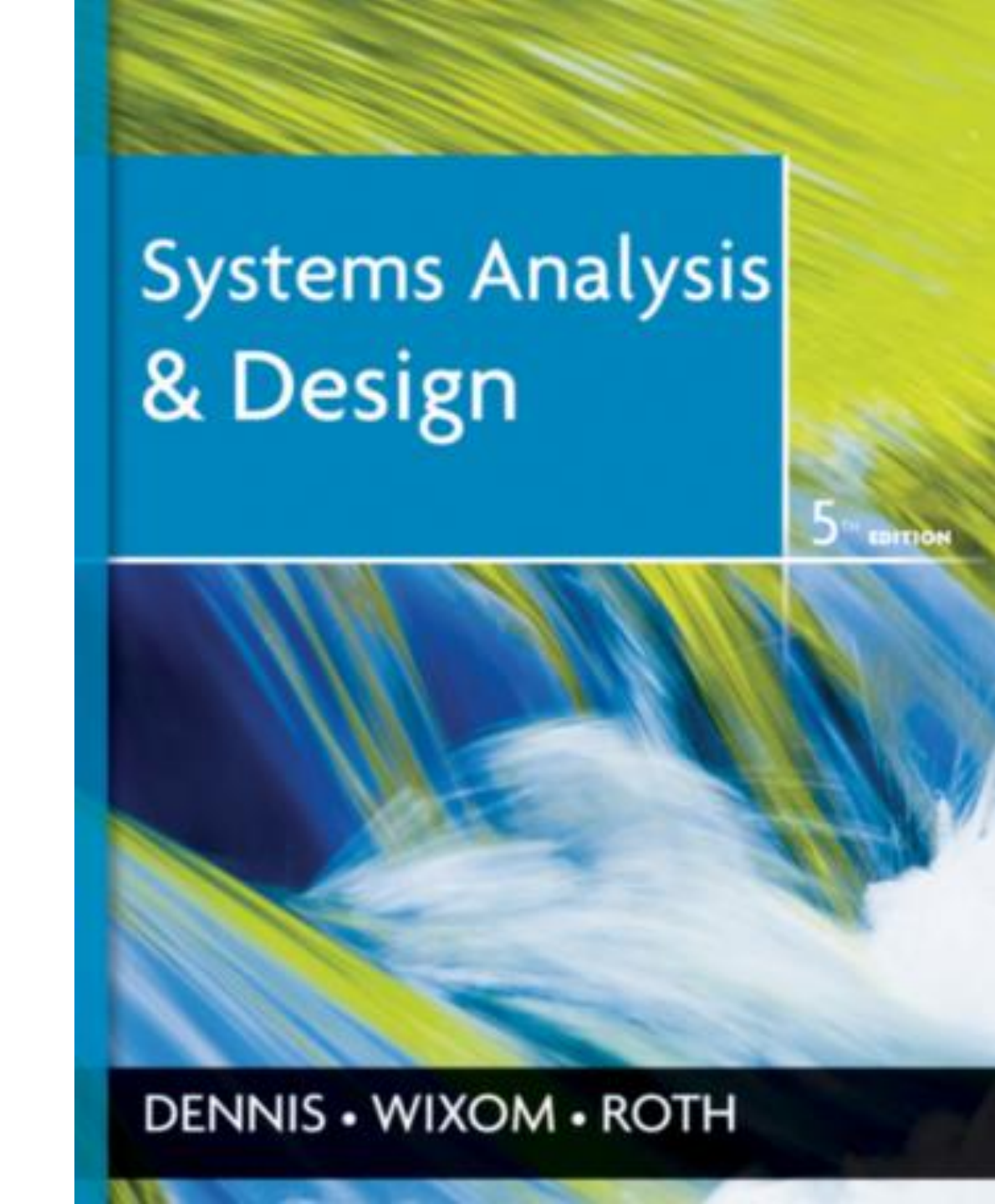




Systems Analysis & Design

5th EDITION

DENNIS • WIXOM • ROTH

System Analysis & Design

Presenter: PhD. Tran Thi Minh Khoa

Agenda

Planning Phase

1. The Systems Analyst & Information Systems Development
2. Project Selection & Management

Design Phase

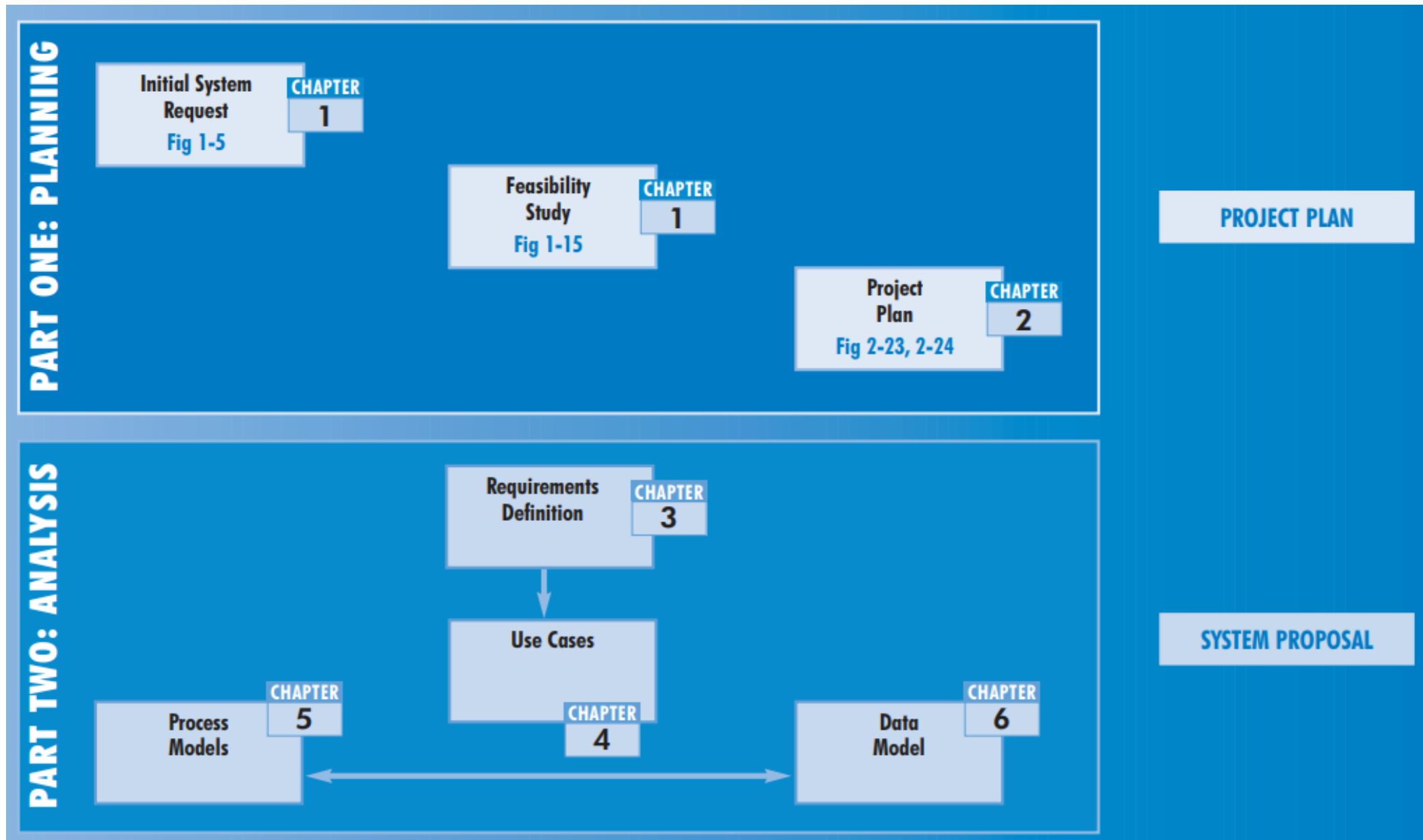
7. Moving Into Design
8. Architecture Design
9. User Interface Design
10. Program Design
11. Data Storage Design

Analysis Phase

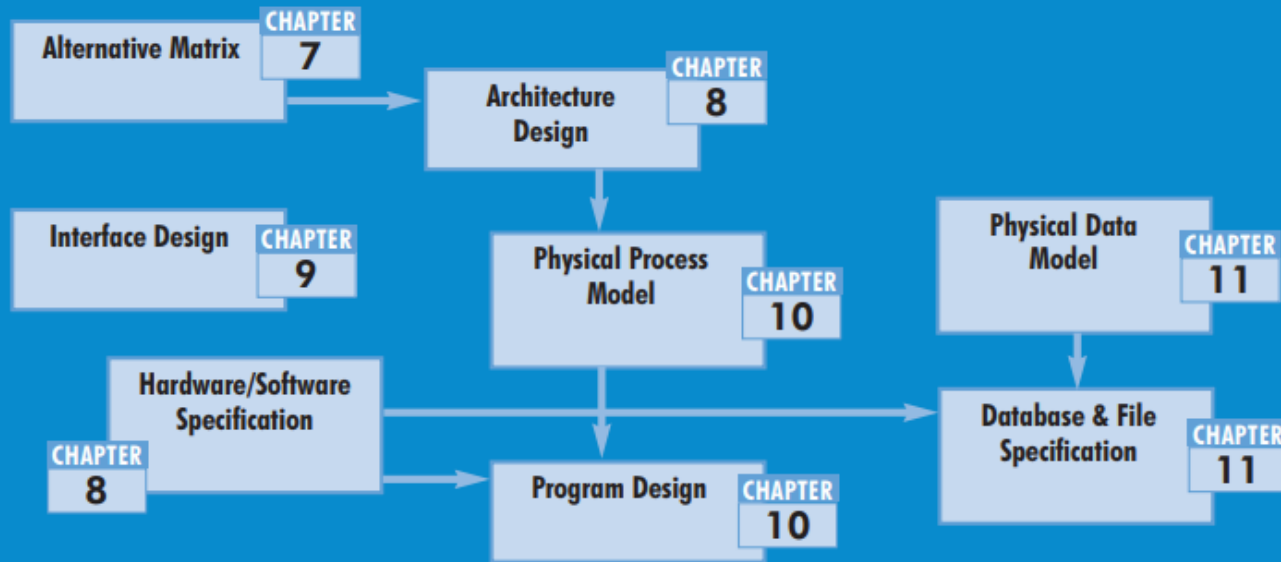
3. Requirement Determination
4. Use Case Analysis
5. Process Modeling
6. Data Modeling

Implementation Phase

12. Moving Into Implementation
13. Transition To The New System
14. The Movement To Object

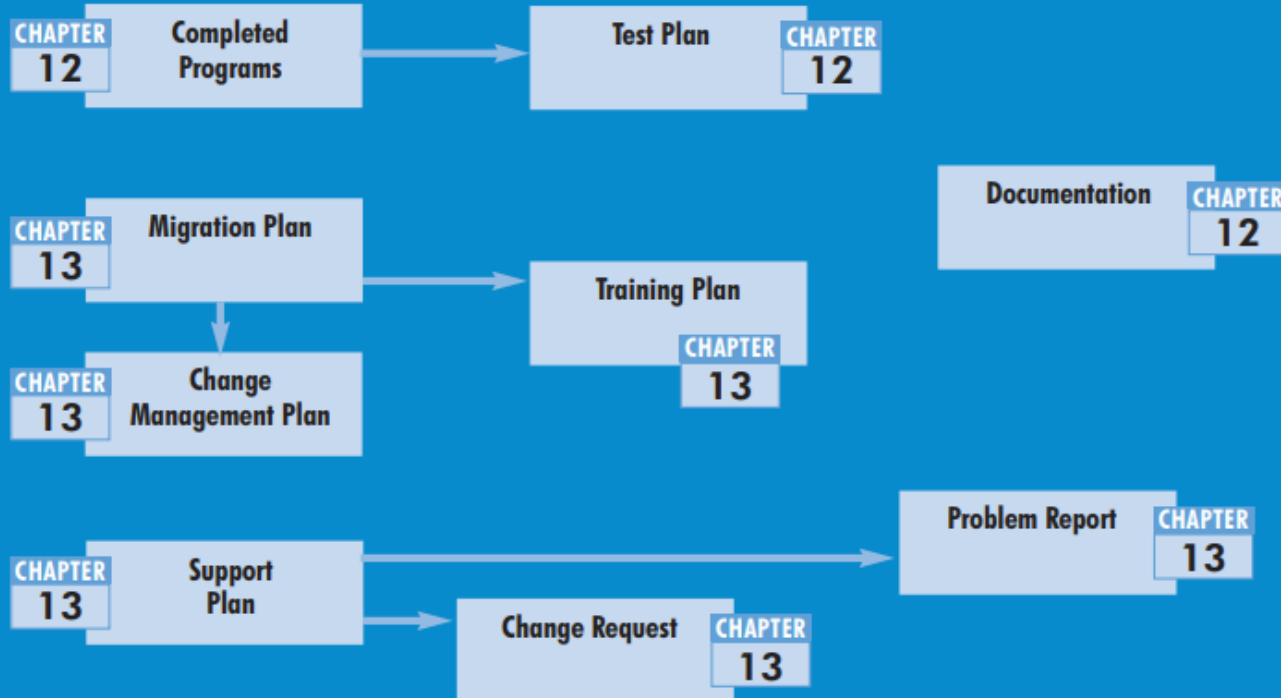


PART THREE: DESIGN



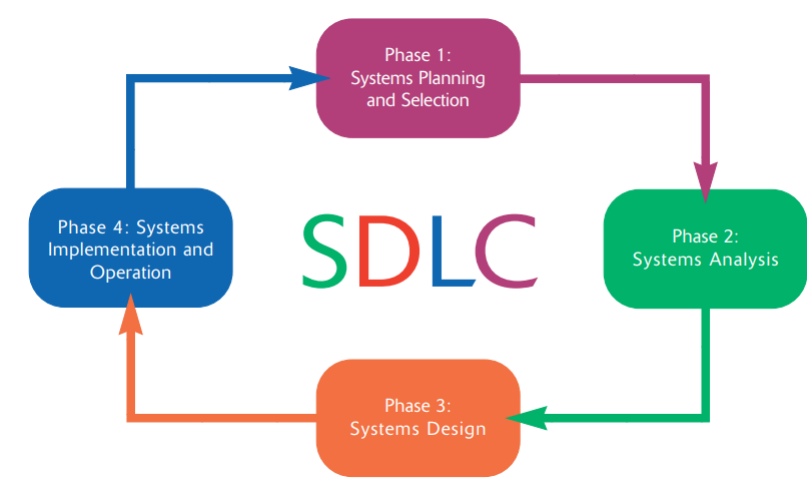
SYSTEM SPECIFICATION

PART FOUR: IMPLEMENTATION



INSTALLED SYSTEM

System Development Life Cycle (SDLC)



The Planning Phase is the fundamental two-step process of understanding why an information system should be developed and creating a plan for how the project team will develop it.

The Design phase decides how the system will operate. This collection of deliverables is the system specification that is handed to the programming team or implementation.

The Analysis phase answers the questions of who will use the system, what the system will do, and where and when it will be used.

The final phase in the SDLC is the **Implementation phase**, during which the system is actually built (or purchased, in the case of a packaged software design)

Chapter 2: PROJECT SELECTION & MANAGEMENT

OBJECTIVES

- Explain how projects are selected in some organizations.
- Describe various approaches to the SDLC that can be used to structure a development project.
- Explain how to select a project methodology based on project characteristics.
- Become familiar with project estimation.
- Be able to create a project work plan.
- Describe project staffing issues and concerns.
- Describe and apply techniques to coordinate and manage the project.
- Explain how to manage risk on the project.

CHAPTER OUTLINE

Introduction

Project Selection

Applying the Concept at Tune Source

Creating the Project Plan

Project Methodology Options

Selecting the Appropriate

Methodology

Estimating the Project Time Frame

Developing the Work Plan

Staffing the Project

Coordinating Project Activities

Managing and Controlling the Project

Refining Estimates

Managing Scope

Timeboxing

Managing Risk

Applying the Concepts at Tune Source

Summary

Appendix 2A—The Function Point Approach

Appendix 2B—Project Management Tools: The Gantt Chart and PERT Chart

Project Selection

Ways to Classify Projects

Size

What is the size? How many people are needed to work on the project?

Cost

How much will the project cost the organization?

Purpose

What is the purpose of the project? Is it meant to improve the technical infrastructure? support a current business strategy? improve operations? demonstrate a new innovation?

Length

How long will the project take before completion? How much time will go by before value is delivered to the business?

Risk

How likely is it that the project will succeed or fail?

Scope

How much of the organization is affected by the system? a department? a division? the entire corporation?

Economic Value

How much money does the organization expect to receive in return for the amount the project costs?

Creating the Project Plan

Once the project is launched by being selected by the approval committee, it is time to carefully plan the project

Project Methodology Options

- **Waterfall Development, 2 variants**
 - **Parallel development method** (evolved to address the lengthy time frame of waterfall development)
 - **V-model** (pays more explicit attention to testing)
- **Rapid Application Development (RAD), 3 variants**
 - **Iterative development**
 - **System prototyping**
 - **Throwaway prototyping**
- **Agile Development**
 - **Extreme programming**

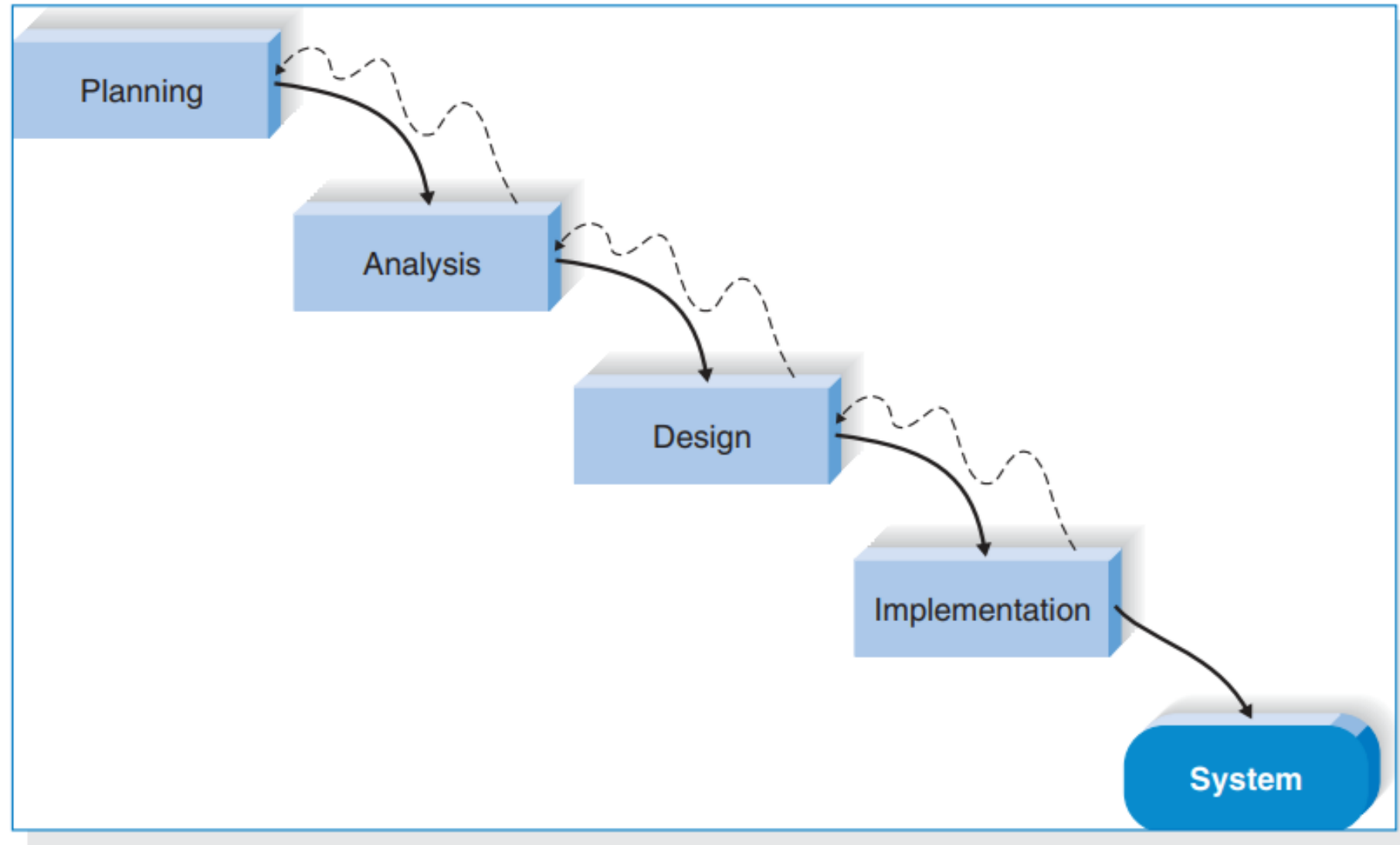
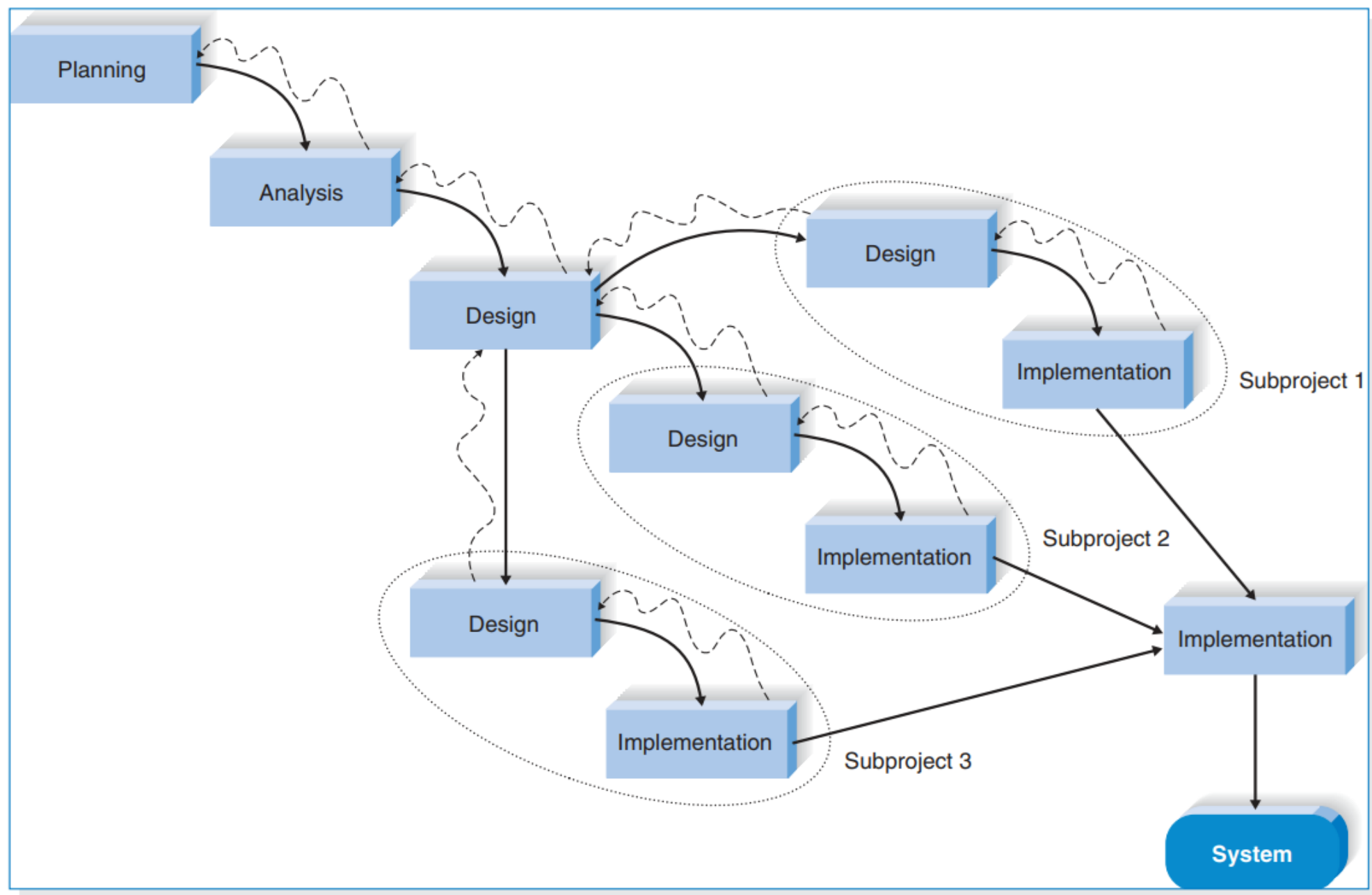


FIGURE 2-2
Waterfall Development



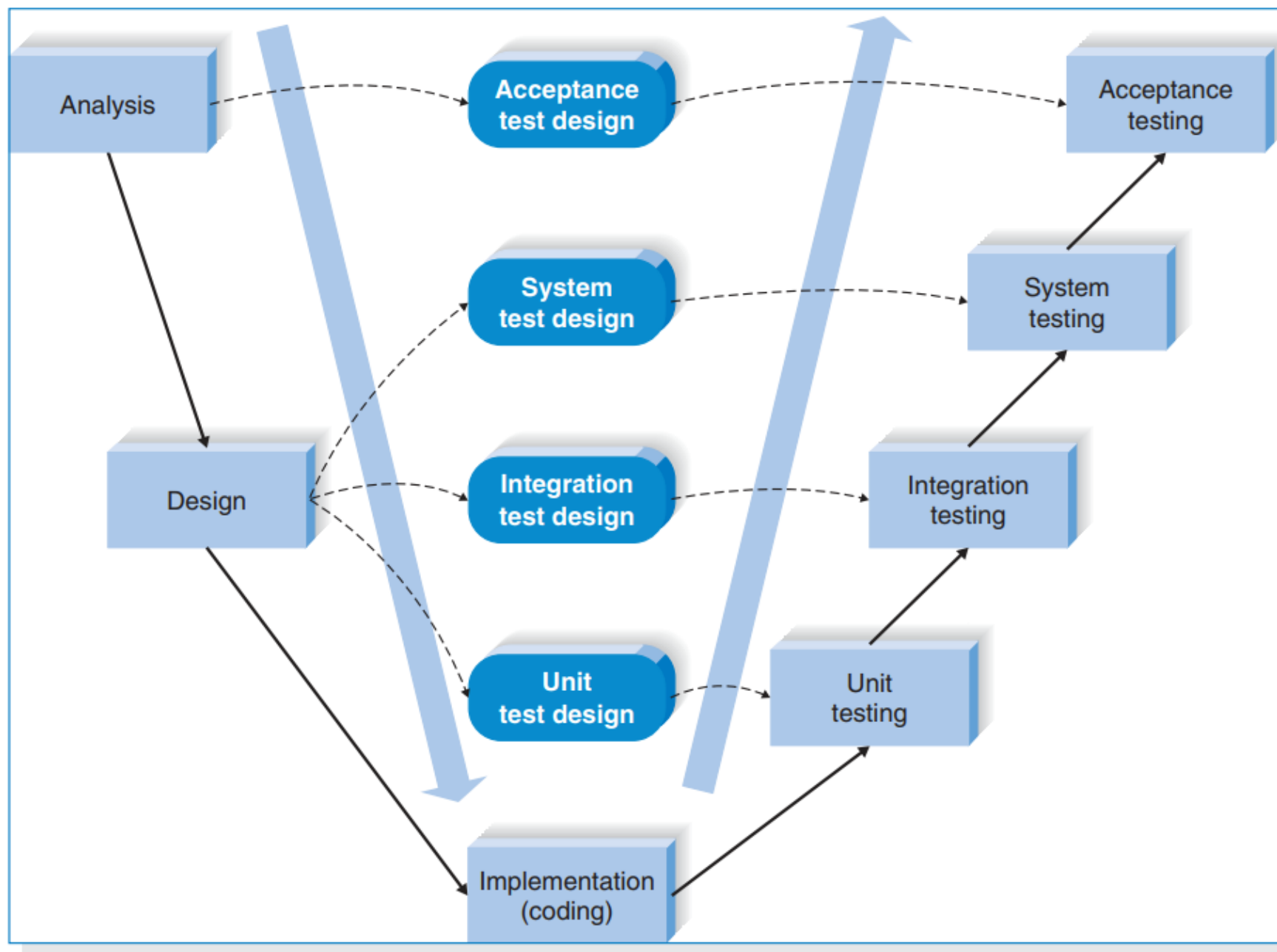


FIGURE 2-4
V-Model

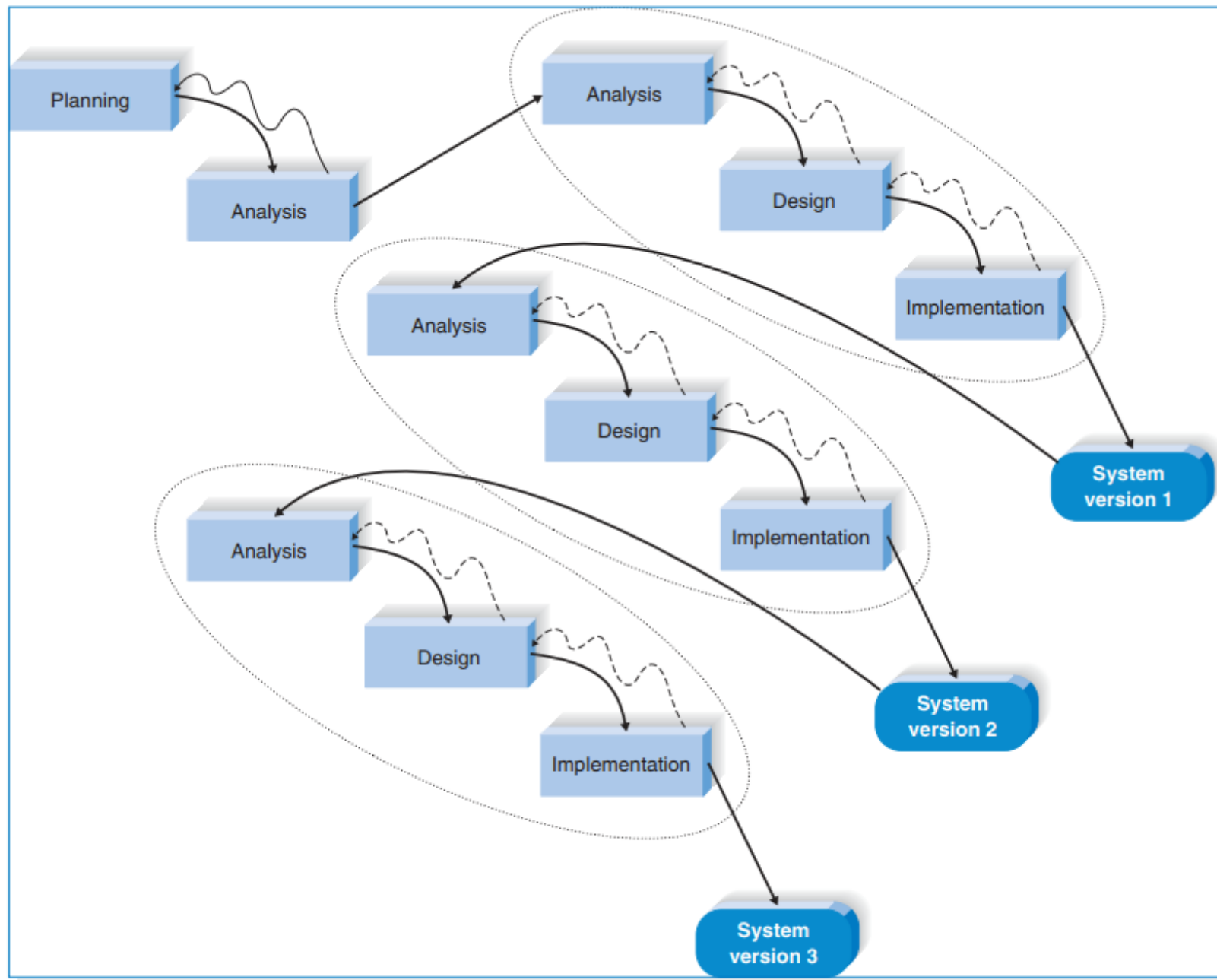
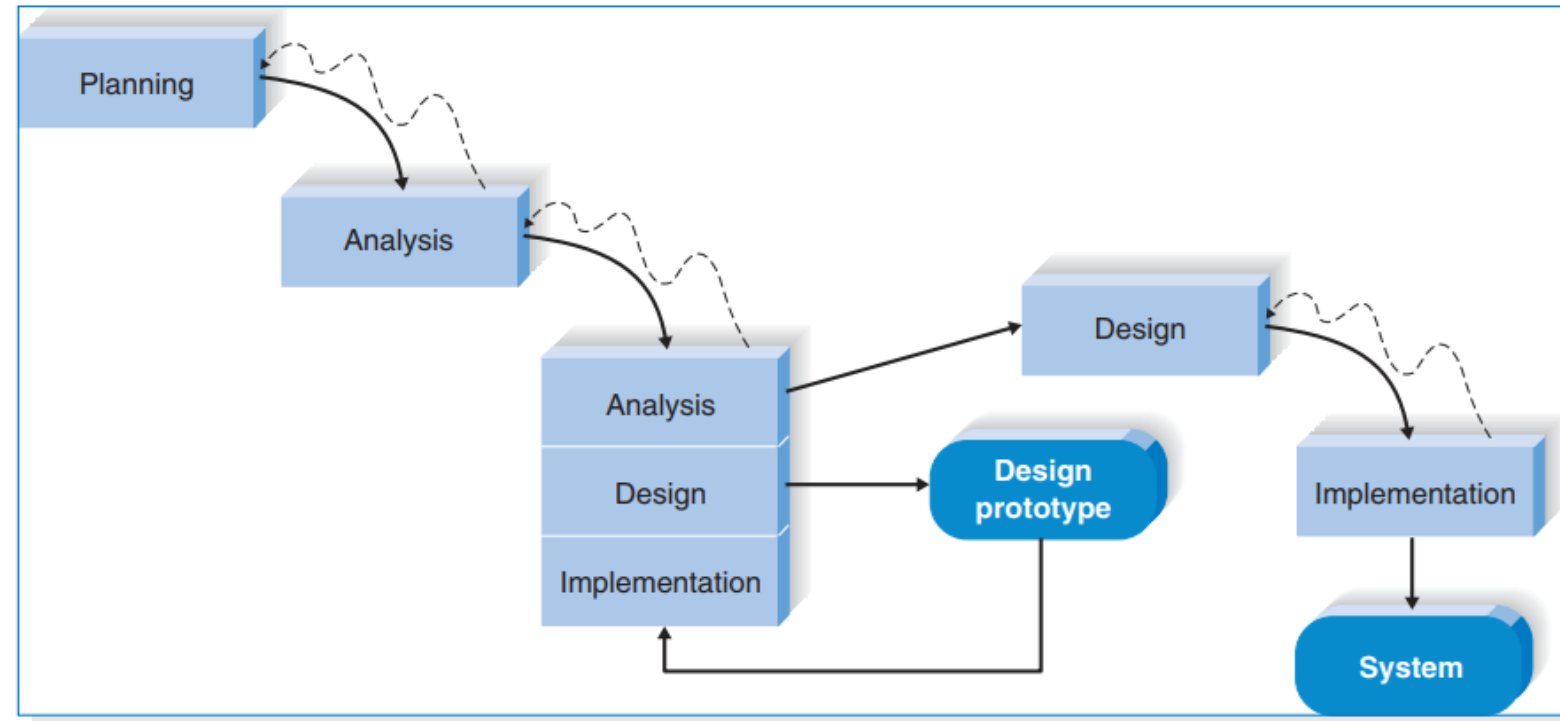
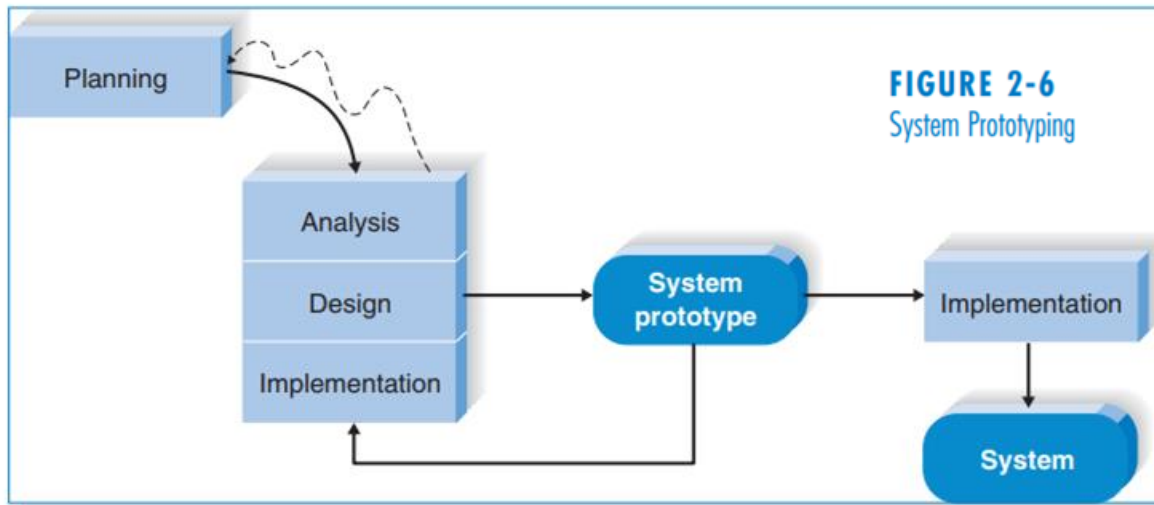


FIGURE 2-5
Iterative Development



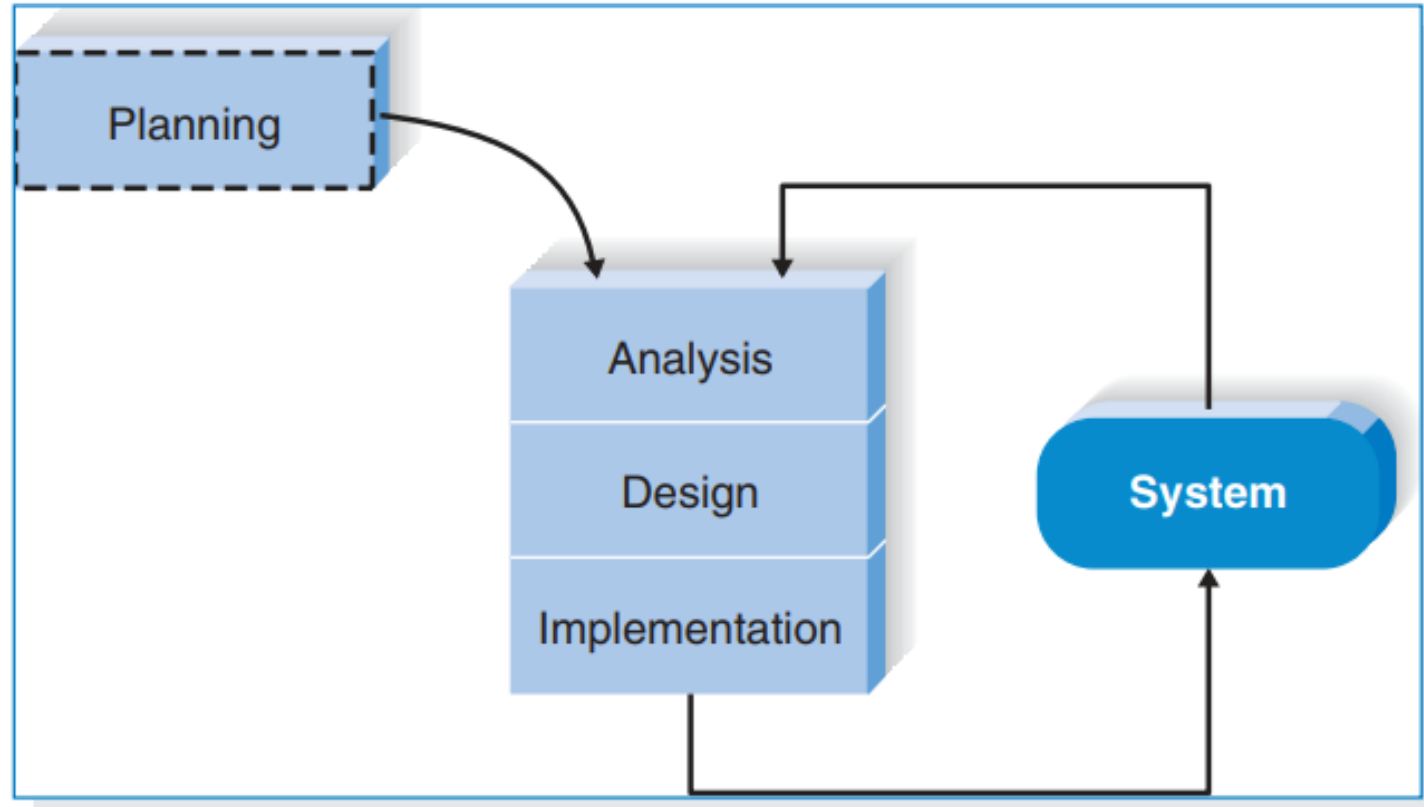


FIGURE 2-8
Extreme Programming

Selecting the Appropriate Development Methodology

Usefulness in Developing Systems	Waterfall	Parallel	V-Model	Iterative	System Prototyping	Throwaway Prototyping	Agile Development
with unclear user requirements	Poor	Poor	Poor	Good	Excellent	Excellent	Excellent
with unfamiliar technology	Poor	Poor	Poor	Good	Poor	Excellent	Poor
that are complex	Good	Good	Good	Good	Poor	Excellent	Poor
that are reliable	Good	Good	Excellent	Good	Poor	Excellent	Good
with short time schedule	Poor	Good	Poor	Excellent	Excellent	Good	Excellent
with schedule visibility	Poor	Poor	Poor	Excellent	Excellent	Good	Good

FIGURE 2-9

Criteria for Selecting a Methodology

Estimating the Project Time Frame

- Estimation is the process of assigning projected values for time and effort.
- Estimation can be performed manually or with the help of an estimation software package like Construx Estimate,™ Costar,™ or KnowledgePLAN® — there are over 50 available on the market
- There are two basic ways to estimate the time required to build a system.
 - The simplest method uses the amount of time spent in the planning phase to predict the time required for the entire project
 - A more precise approach to estimation is called the function point approach

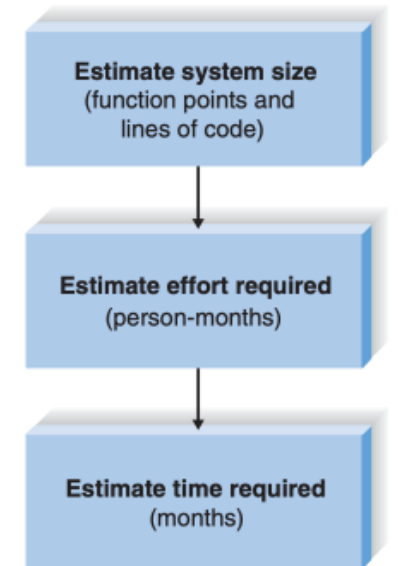
	Planning	Analysis	Design	Implementation
Typical industry standards for business applications	15%	20%	35%	30%
Estimates based on actual figures for first stage of SDLC	Actual: 4 person-months	Estimated: 5.33 person-months	Estimated: 9.33 person-months	Estimated: 8 person-months

SDLC = systems development life cycle.

Estimating Project Time Using Industry Standards

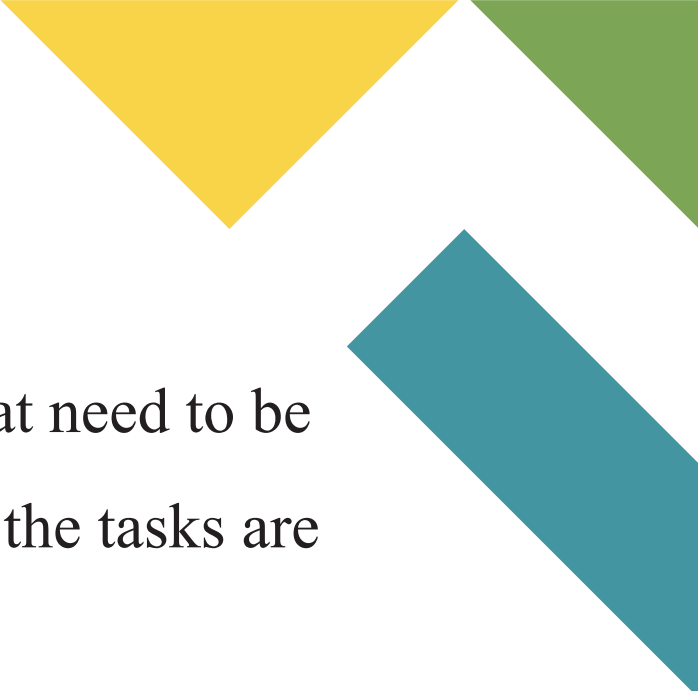
Task Information	Example
Name of the task	Perform economic feasibility
Start date	Jan 05, 2013
Completion date	Jan 19, 2013
Person assigned to the task	Project sponsor Mary Smith
Deliverable(s)	Cost-benefit analysis
Completion status	Complete
Priority	High
Resources needed	Spreadsheet software
Estimated time	16 hours
Actual time	14.5 hours

Task Information



Estimating Project Time, Using the Function Point Approach

Developing the Work Plan



To create a work plan, the project manager identifies the tasks that need to be accomplished and determines how long each one will take. Then the tasks are organized within a work breakdown structure.

- Identify Tasks * (Fig. 2-12)
- The project work plan (Fig. 2-13)

Staffing the Project

Staffing the project includes determining how many people should be assigned to the project, matching people's skills with the needs of the project, motivating them to meet the project's objectives, and minimizing project team conflict that will occur over time

Staffing Plan

1. The first step to staffing is determining the average number of staff needed for the project
2. Motivation : project managers need to motivate the people to make the project a success
3. Handling Conflict : The third component of staffing is organizing the project to minimize conflict among group members



Coordinating Project Activities

1. CASE (*Computer-aided software engineering*) Tools
2. Standards
3. Documentation

Managing and Controlling the Project



1. Refining Estimates
2. Managing Scope
3. Timeboxing
4. Management Risk